RKG Wallah Academy

Class 6 Science — Complete Notes

NCERT based chapter-wise notes covering Biology, Physics and Chemistry basics.

Note: Ye NCERT-based summary notes hain. Diagrams NCERT se apni copy mein banao aur exam se pehle apni book se ek baar mila lo.

1. Food: Where Does It Come From?

• Plants give us cereals, pulses, fruits, vegetables, spices and oils. Animals give milk, eggs, meat, honey.

• Plant parts we eat: root (carrot), stem (potato, sugarcane), leaf (spinach), flower (cauliflower), fruit (tomato), seed (rice).

• Herbivores eat plants; carnivores eat animals; omnivores eat both.

2. Components of Food

Nutrient	Function	Sources
Carbohydrates	Energy	Rice, wheat, potato, sugar
Proteins	Growth and repair of body (body-building)	Pulses, milk, eggs, fish, soya
Fats	Energy (more than carbohydrates), insulation	Butter, oil, ghee, nuts
Vitamins	Protective food — keep body healthy	Fruits, vegetables, milk
Minerals	Healthy bones, blood	Salt, milk, green vegetables
Water	Digestion, transport; 70% of body	Drinks, fruits
Roughage (fibre)	Helps removal of waste	Vegetables, fruits, whole grains

• Test for starch: iodine solution turns blue-black. Protein: copper sulphate + caustic soda gives violet. Fat: brown paper shows oily patch.

• Balanced diet: contains all nutrients in right amounts.

• Deficiency diseases: Vitamin A – night blindness; Vitamin C – scurvy; Vitamin D – rickets; Vitamin B1 – beriberi; Calcium – tooth decay and weak bones; Iron – anaemia; Iodine – goitre; Protein – kwashiorkor.

3. Fibre to Fabric

• Plant fibres: cotton (from cotton bolls), jute (stem). Animal fibres: wool (sheep, yak), silk (silkworm cocoon).

• Yarn is made from fibre by spinning (using a spindle/takli or charkha). Fabric is made by weaving or knitting.

• Cotton: ginning separates seeds from fibre. Silk: sericulture is rearing of silkworms.

4. Sorting Materials into Groups

• Objects are grouped by properties: appearance (lustre), hardness, solubility, float/sink, transparency.

• Transparent (glass, clear water), translucent (butter paper) and opaque (wood, metal) materials.

• Materials soluble in water (sugar, salt) and insoluble (sand, chalk).

5. Separation of Substances

Method	Used for
Handpicking	Large impurities in grains
Winnowing	Separating husk from grain by wind
Sieving	Separating different-sized particles (flour and bran)
Sedimentation, decantation	Insoluble heavier particles settle; pour off liquid
Filtration	Insoluble solid from a liquid
Evaporation	Getting salt from sea water
Condensation	Getting water from vapour
Magnetic separation	Iron from other materials

- Saturated solution: cannot dissolve any more solute at that temperature.

6. Changes Around Us

• **Reversible changes:** can be reversed (melting wax, folding paper, stretching rubber band).

• **Irreversible changes:** cannot be reversed (cooking food, burning wood, rusting, curdling milk).

• Changes can be caused by heating, cooling, mixing or applying force. Expansion and contraction on heating/cooling.

7. Getting to Know Plants

• Herbs (small, soft stem), shrubs (branched, hard stems), trees (tall, thick stem).

• Parts: **root** (absorbs water and minerals, anchors), **stem** (conducts water, holds leaves), **leaf** (makes food by photosynthesis; transpiration), **flower** (reproduction).

• Taproot (dicot, e.g. mustard) and fibrous root (monocot, e.g. wheat). Reticulate venation (dicot) and parallel venation (monocot).

• Flower: sepals, petals, stamens (male: anther + filament), pistil/carpel (female: stigma, style, ovary).

• Photosynthesis: Chlorophyll + sunlight + CO_2 + water → glucose + oxygen.

8. Body Movements

• Human skeleton has 206 bones. Joints: ball and socket (shoulder, hip), pivotal (neck), hinge (knee, elbow), fixed (skull).

• Skull protects brain; rib cage protects heart and lungs; backbone (vertebral column) protects the spinal cord.

• Muscles work in pairs (contract and relax) to move bones. Animals: earthworm moves by muscles and bristles; snail by muscular foot; fish by fins; birds by wings; snake by body muscles.

9. The Living Organisms and Their Surroundings

• **Habitat**: surroundings where an organism lives. **Adaptation:** features that help it survive.

• Terrestrial habitats: forests, grasslands, deserts, mountains. Aquatic habitats: ponds, lakes, rivers, seas.

• Desert plants (cactus) have spines and thick stem; camel has long legs and stores fat in hump. Fish have streamlined body and gills; frog is amphibian.

• Biotic components: living things. Abiotic: air, water, soil, light, temperature.

• Characteristics of living things: need food, respire, respond to stimuli, reproduce, grow, excrete, die.

10. Motion and Measurement of Distances

• Standard unit of length: metre (m). 1 km = 1000 m; 1 m = 100 cm; 1 cm = 10 mm.

• Measure with a scale/tape: keep the eye directly above the mark, start from the zero mark.

• Types of motion: rectilinear (straight), circular, periodic (repeats after fixed time, e.g. pendulum), rotational, oscillatory.

• Curved lines are measured using a thread.

11. Light, Shadows and Reflections

• Luminous objects give out their own light (Sun, stars, bulb); non-luminous reflect light (moon).
• Transparent, translucent and opaque objects. Light travels in straight line.
• Shadow forms when an opaque object blocks light. Needs a source, an opaque object, and a screen. Pinhole camera forms an inverted image.
• Mirror reflects light and forms images.

12. Electricity and Circuits

• Cell has positive and negative terminals. Bulb glows when a closed circuit is made.

• A switch opens or closes a circuit. Conductors (metals, graphite) allow current; insulators (rubber, plastic, wood, glass) do not.

• Safety: never touch electrical switches with wet hands.

13. Fun with Magnets

• Magnetic materials: iron, nickel, cobalt. Non-magnetic: plastic, wood, paper.

• Magnet has two poles (N, S). Like poles repel; unlike poles attract. Freely suspended magnet points north–south (basis of compass).

• Magnet poles are strongest at ends. Keep magnets with keeper pieces; avoid heating or hammering.

14. Water

• Water cycle: evaporation → condensation (clouds) → precipitation → collection (rivers, seas, groundwater).

• Water is used for drinking, agriculture, industry. Conserve water: rain-water harvesting, drip irrigation, fix leaks.

• Droughts and floods are extremes of water availability.

15. Air Around Us

• Composition of air: nitrogen ≈ 78%, oxygen ≈ 21%, argon, CO_2 and others ≈ 1%.

• Oxygen supports burning and respiration; plants use CO_2; nitrogen is needed by plants (nitrogen fixation).

• Air occupies space, has weight, moves as wind and exerts pressure.

16. Garbage In, Garbage Out

• Waste: biodegradable (kitchen waste) and non-biodegradable (plastic, glass).

• Composting, vermicomposting (earthworms), recycling paper and plastic, 3 Rs: **Reduce, Reuse, Recycle**.